SEC 4 · LEC 4 · NUCLEAR THERMAL ROCKETS

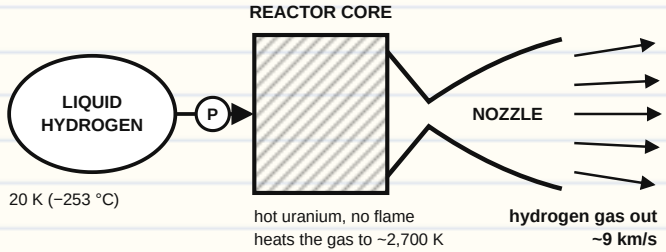
A kettle with no fire

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

S4 / L4 — Nuclear thermal rockets **date · pg _**

1 · NUCLEAR THERMAL ROCKET

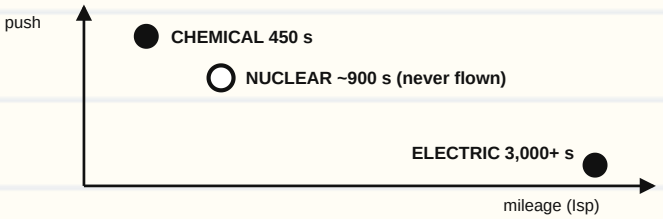


2 · WHY THE MILEAGE DOUBLES

CHEMICAL exhaust	NUCLEAR exhaust
18	2
steam: heavy	hydrogen: light
3,500 K	2,700 K (cooler!)
4.4 km/s · 450 s	8.8 km/s · 900 s

light molecules fly faster: $v_e \propto \sqrt{(T \div M)}$

3 · THE THREE FAMILIES



✏️ **PENCIL KEY** · hatched box = reactor core · Ⓟ = pump · number in circle = how heavy the exhaust molecule is · ● flown · ○ never flown

⭐ **3 TAKEAWAYS** 3 lines

1. A reactor, not a fire, heats hydrogen; the nozzle does the rest.
2. Light exhaust is fast exhaust: Isp ~900 s, twice chemical, with real thrust.
3. Ground-tested in the 1960s, never flown: cost, politics, radiation.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
What heats the propellant in a nuclear thermal rocket?	nuclear thermal rocket (NTR): a reactor heats the propellant; nothing burns
Path of the hydrogen, tank to exhaust?	liquid hydrogen → pump → reactor (heated, ~2,700 K) → nozzle (expands, pressure ↓, speed ↑) → thrust
Isp compared with chemical? Why?	Isp ≈ 2× chemical (~900 s vs 450 s) · and still high thrust
What sets the temperature limit?	why: exhaust is light hydrogen, not heavy steam (not because it is hotter)
Why is it attractive for Mars?	limit = what the reactor materials can stand
What is payload?	2× Isp → about half the mass for the same trip, or 2–3× the payload
Low-enriched uranium: what and why?	payload = the useful cargo: instruments, cameras, crew
Why has it never flown?	Mars: faster trip → less radiation exposure for the crew
The two examples: what are they really?	low-enriched uranium (LEU): safer to build and test, fewer security rules, cheaper
? is it hotter than a chemical rocket	ground-tested 1960s · politics + cost → never flown
	theory: fuel mixed into the fluid (liquid / gas core) → hotter → higher Isp
	examples: bimodal NTR (drawing only) · Project Orion (really nuclear pulse: bombs)

SEC 4 · LEC 4 · TEACH

Not hotter. Lighter.

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

A normal rocket heats its gas by burning it. A nuclear rocket is a kettle with no fire. Inside is a block of special metal, uranium, that gets scorching hot all by itself when its atoms split. The engine squirts the lightest gas there is, hydrogen, through holes in that hot block. The gas comes out blazing and shoots through the same bell nozzle as before.

Why is that better? Throw a ping-pong ball and a bowling ball with the same arm: the ping-pong ball leaves much faster. The fire in a chemical rocket makes steam, which is the bowling ball. A reactor doesn't need to burn anything, so it can throw plain hydrogen, the ping-pong ball. Same kind of heat, twice the exhaust speed, twice the mileage, and it still pushes hard. The catch: the hot block is radioactive, the engine is heavy, and nobody has ever flown one. They ran on test stands in the Nevada desert in the 1960s.

🏷 The official names (name first, then what it means)

- **Nuclear thermal propulsion, NTP** (**nuclear thermal rocket, NTR**; solid-core nuclear rocket): a fission reactor heats a propellant that leaves through a nozzle.
- **Nuclear fission**: a heavy atom (uranium-235) splitting in two and releasing heat. A **reactor** keeps that going at a controlled rate. A **fuel element** is one of the rods that holds the uranium; together they are the **reactor core**.
- **Working fluid** (here the same thing as the propellant): the hydrogen that gets heated and thrown. Three different words now: **nuclear fuel** is the uranium, **propellant** is the hydrogen, and nothing is "fuel" in the burning sense.
- **Molecular mass, M** (molar mass): how heavy one molecule of the exhaust is. Hydrogen 2, steam 18.
- **Enrichment**: the share of uranium that is the splittable kind, U-235. **Low-enriched uranium, LEU**: under 20%. **Highly enriched, HEU**: 20% and up; weapons grade is about 90%.
- **Payload**: the part of the spacecraft that is the reason for the trip. **Payload fraction**: its share of the total mass.
- **Bimodal nuclear thermal rocket**: one reactor, two jobs. Thrust during burns, electricity the rest of the time.
- **Liquid-core / gas-core nuclear rocket**: the slide's "mixing nuclear fuel and the working fluid". Paper designs only.
- **Nuclear pulse propulsion** (Project Orion): riding a string of small nuclear explosions. Not a thermal rocket.
- **Radiation exposure** (dose): how much damaging radiation a crew absorbs. A **shadow shield** sits between reactor and crew.

🛠 How it actually works

Step	The mechanism, with numbers (the 1960s NERVA engine)
1 · The heat source	Splitting 1 kg of uranium-235 releases about 80 million MJ. Burning 1 kg of hydrogen and oxygen releases 13. Six million times more, so energy is never the limit.
2 · Flow path	Liquid hydrogen at 20 K → turbopump → channels in the nozzle wall (cooling it) → down through hundreds of holes in the hot core → out as gas at about **2,700 K** → throat → bell.
3 · Why it is faster	Exhaust speed goes as √(T ÷ M). Chemical: 3,500 K ÷ 13.6 (steam with spare hydrogen). Nuclear: 2,700 K ÷ 2. Ratio of the square roots ≈ 2.3. The reactor is *cooler* and still wins, because its exhaust is seven times lighter.
4 · The two numbers	$v_e \approx 8.3$ km/s, so Isp = 8,300 ÷ 9.81 ≈ **850 s** (NERVA measured 841 s). Thrust **246,000 N**, about 25 tonnes of push.
5 · Check it	$\dot{m} = F \div v_e = 246{,}000 \div 8{,}300 \approx 30$ kg of hydrogen per second. Jet power = $\frac{1}{2}\dot{m}v_e^2 \approx 1{,}000$ MW. The reactor was rated 1,100 MW. It adds up.
6 · What it buys	Leaving Earth orbit for Mars needs about 4 km/s. Rocket equation: at 450 s the ship is 60% propellant; at 900 s it is 36%. Spend the saving on payload, or on speed for a shorter trip.
7 · The limits	The core must stay solid, so about 2,700–2,900 K at most. Hot hydrogen eats the fuel rods. Liquid hydrogen is 14 times less dense than water: huge tanks. Reactor and shield weigh tonnes: thrust is only 3–4 times the engine's weight, against 70–180 for chemical.
8 · History	US Project Rover and NERVA, 1955–1973: about 20 reactors run in Nevada. In 1969 one engine was started 28 times. Cancelled when the Mars plans were. Never flown.

SEC 4 · LEC 4 · FACT CHECK

Three real errors in this lecture

This lecture has more wrong statements than any so far. The first three change what you would say to a friend.

⛔ Wrong, and worth getting right

- **"Very high specific impulse since the working temperatures are higher."** A nuclear core runs *cooler* than a chemical flame (2,700 K against 3,500 K), because the fuel rods must not melt. The Isp doubles because the exhaust is light hydrogen and not heavy steam. He half-remembers the rule ("the square root of the temperature"). The full rule is √(temperature ÷ molecular mass).
- **"NASA's ORION nuclear project… developed in 1986… nuclear thermal rockets."** Three errors. Project Orion ran **1958–1965**, led by a company (General Atomics) for the US military. It is **not a thermal rocket**: the ship drops small nuclear bombs behind itself and rides the blasts on a giant plate with shock absorbers. That is exactly what his own slide diagram labels: pusher, shock absorbers, propellant magazines. The 1963 treaty banning nuclear explosions in space killed it. And it has nothing to do with today's Orion crew capsule.
- **"It is also attractive for interstellar flight."** No. A nuclear thermal rocket gives perhaps 10–20 km/s. The nearest star is 40 trillion km away: at Voyager's 17 km/s that is **75,000 years**. It is an engine for reaching Mars, not other stars.

🔎 Caption errors

Captions said	Instructor meant
"nuclear them a rocket / rock edge"	nuclear **thermal rocket(s)**
"on the QM state of the art… react to development"	the **current** state of the art… **reactor** development
"specific import" · "lift of mass" · "tripped"	specific **impulse** · **lift-off** mass · **tripled**
"radiation explosion"	radiation **exposure** (slide: "exposition")
"hitting some kind of liquid"	**heating** some kind of liquid
"knotholes"	**nozzles**
"Nysa… by mode on nuclear rocket"	**NASA**… **bimodal** nuclear rocket
"NASA's Arrian nuclear project"	the **Orion** nuclear project
"nuclear production / proportion"	nuclear **propulsion**

▲ Where else he was loose

- **"First proposed and sketched in the 1960s… some prototypes."** Proposed in the 1940s. By the 1960s the US had built and run about 20 reactors at full power. It was far more than sketched.
- **"Right now it's 2018."** The lecture is eight years old. Since then NASA and DARPA started a flight demo (DRACO, 2023) and cancelled it in 2025, as far as I know.
- **"The required lift-off mass is half."** Nobody plans to lift off from the ground on one: too heavy for its thrust, and radioactive. It is a stage you start in orbit. The honest version: the mass you must put *into Earth orbit* for a Mars trip roughly halves.
- **"The liquid at high temperature… high velocity liquid."** It goes in as liquid at 20 K and is a gas long before it reaches 2,700 K. Liquids don't expand through nozzles.
- **"We all know what happens if we cannot withstand some nuclear reaction."** A reactor cannot explode like a bomb. The real failure is fuel rods cracking or melting.
- **"A dispute among the scientific community."** The physics isn't disputed; it was demonstrated. The obstacles are cost, testing (the exhaust is radioactive) and public acceptance.
- **"Low-enriched uranium… promoted in the European Union."** The switch to low-enriched fuel was NASA's own choice in the late 2010s. I don't know of a European programme he could mean.
- **"Mixing nuclear fuel and the working fluid."** These are liquid-core and gas-core designs: let the fuel melt or vaporize so temperature is no longer capped. Isp of 1,500 to 5,000 s on paper. None has been built.

📒 Question bank cards

- A nuclear rocket runs cooler than a chemical one. Why is its Isp double?
- Trace the hydrogen from tank to exhaust.
- Nuclear fuel, propellant, working fluid: which is which here?
- Why would a crew to Mars want one? Give two reasons.
- Project Orion: what was it really, and why did it end?

SEC 4 · LEC 4 · VOCABULARY

Every word used, in plain English

42 entries. "Repeat" = you have met it before.

Nuclear propulsion — using a nuclear reactor (or explosions) as an engine's energy source. *Repeat, L1.*

Nuclear thermal rocket (NTR) = nuclear thermal propulsion (NTP) — a reactor heats hydrogen, which leaves through a nozzle. *Kettle with no fire.*

Nuclear fission — a heavy atom splitting in two and releasing a lot of heat.

Uranium / U-235 — the heavy metal used as nuclear fuel / the kind of uranium atom that splits easily.

Nuclear reactor — a machine that keeps fission going at a steady, controlled rate.

Reactor core / fuel element — the hot heart of the reactor / one of the rods that holds the uranium.

Nuclear fuel — the uranium. It is not burned and not thrown out. *Not the same as rocket fuel.*

Working fluid — the substance that is heated and expanded to do the work. Here hydrogen, which is also the propellant.

Liquid hydrogen (LH_2) — hydrogen chilled to 20 K. Very light, very bulky. *Repeat, L2.*

Efficiency / exceptional efficiency — here: high specific impulse, a lot of Δv per kilo of propellant. *Repeat.*

Specific impulse (Isp) — exhaust speed ÷ 9.81, in seconds. NTR: about 850–900 s. *Repeat.*

Working temperature — how hot the gas gets inside the engine.

Square root (√) — the number that, multiplied by itself, gives the original. √4 = 2.

Molecular mass (M) — how heavy one molecule is, on a scale where hydrogen gas is 2 and water is 18. *Ping-pong ball against bowling ball.*

Proportional to (∝) — 'goes up and down together with'.

Theoretical — worked out on paper, not built.

Liquid-core / gas-core rocket — designs where the uranium itself is molten or gas and mixed with the propellant. Hotter, never built.

Withstand — survive without breaking or melting.

Material properties — what a material can take: melting point, strength, resistance to attack.

Interplanetary / interstellar — between planets / between stars. The second is thousands of times farther.

State of the art — the best that current technology can do.

Prototype / model — a first working version built for testing / a mock-up or a calculation.

Lift-off mass — the total mass of the rocket as it leaves the pad.

Payload — the cargo that is the reason for the mission: instruments, cameras, crew.

Payload fraction — the share of the total mass that is payload. Usually a few percent.

Radiation exposure (dose) — how much harmful radiation a person absorbs. In deep space it comes from the Sun and the galaxy.

Enrichment — raising the share of U-235 in uranium. Natural uranium has 0.7%.

Low-enriched uranium (LEU) — under 20% U-235. Cannot be made into a bomb easily, so fewer security rules.

Highly enriched uranium (HEU) — 20% U-235 and up. The 1960s engines used about 93%.

Security regulations — the legal controls on who may handle bomb-grade material.

Budget / schedule — the money / the timeline.

Turbine pump = turbopump — the pump that forces hydrogen into the reactor. *Repeat, L2.*

Nozzle / expand / pressure — the bell that turns pressure and heat into speed. *Repeat, L2.*

Fluid — anything that flows: liquid or gas. The exhaust here is a gas. *Repeat.*

Bimodal — two modes: the same reactor gives thrust and electricity.

Artist's concept — a painting of something that does not exist yet.

Nuclear pulse propulsion — propelling a ship with a series of nuclear explosions behind a pusher plate. *Project Orion.*

Pusher plate / shock absorber — the thick plate that takes each blast / the springs that smooth the kick.

Shadow shield — a slab between the reactor and the crew that blocks its radiation.

Project Rover / NERVA — the US nuclear rocket programme, 1955–1973. *Who's Who.*

Project Orion (1958–1965) — the US nuclear pulse study. Not the Orion crew capsule. *Who's Who.*

DRACO — the cancelled NASA and DARPA flight demo of the 2020s. *Who's Who.*

SEC 4 · LEC 4 · COPY INTO NOTES

Where each piece goes in the notebook

Routes A to D for this lecture. Section 4 ends here, so route B is now the full sentence for the Section 4 map page.

A · Equations & key numbers page (p3–4)

```
vₑ ∝ √(T ÷ M)      hot + LIGHT exhaust is fast
                   chemical 3,500 K ÷ 13.6 · nuclear 2,700 K ÷ 2
                   → about 2× the exhaust speed
nuclear thermal    Isp ~850–900 s · 246,000 N · 30 kg/s H₂
                   1,100 MW reactor (NERVA, 1960s)
Mars departure     4 km/s: 60% propellant at 450 s
                           36% propellant at 900 s
```

B · Section 4 map page (write it now, from memory first)

```
S4 · propulsion: throw mass back → F = ṁ × vₑ
S4 · chemical: burn → throat → bell; big ṁ, Isp 250–450 s
S4 · electric: ionize → grids → neutralize; mg/s, Isp 3,000+ s
S4 · nuclear: reactor heats H₂ (light = fast); Isp ~900 s,
     never flown
S4 · the trade: push vs mileage; energy from propellant /
     solar panels / reactor
```

D · Question bank ("but how?" cards)

```
Q: A nuclear core is cooler than a flame. Why is Isp double?
A: vₑ ∝ √(T ÷ M). Exhaust is H₂ (mass 2), not steam (18).
Q: How does the hydrogen get hot with nothing burning?
A: It flows through holes in a core heated by uranium fission.
Q: Why not launch from the ground with it?
A: Engine too heavy for its thrust (3–4× its weight) and
   radioactive. It is started in orbit.
Q: What was Project Orion?
A: Nuclear PULSE: bombs behind a pusher plate. 1958–1965.
```

C · One-line word bank (new lines)

```
p__ · nuclear thermal rocket = NTR / NTP · reactor heats H₂  → C22
p__ · fission · heavy atom splits, releases heat
p__ · reactor / reactor core · controlled fission / its hot heart
p__ · fuel element · rod that holds the uranium
p__ · nuclear fuel vs propellant · uranium (stays) vs H₂ (thrown)
p__ · working fluid · what gets heated + expanded: H₂
p__ · molecular mass (M) · how heavy one molecule is  → C22
p__ · payload · the cargo the trip is for  → C23
p__ · payload fraction · payload ÷ total mass
p__ · lift-off mass · total mass leaving the pad
p__ · enrichment · share of U-235 in the uranium  → C23
p__ · LEU / HEU · under 20% / 20% and up (bomb-grade ~90%)
p__ · radiation exposure = dose · radiation a crew absorbs
p__ · shadow shield · slab between reactor and crew
p__ · bimodal NTR · one reactor: thrust + electricity
p__ · liquid-core / gas-core · molten or gas fuel; paper only
p__ · nuclear pulse propulsion · ride bomb blasts (Orion)
p__ · interstellar · between stars; NOT reachable this way
```

Fix one earlier card: in Lecture 1's question bank I wrote "Same heat, lighter exhaust" for nuclear. Change "Same heat" to "**Cooler than a flame, but**". Lecture 1's PDF is reissued as v2 with that corrected.

SEC 4 · LEC 4 · CONCEPT CARDS

Route E: pages C22 and C23

Two notebook pages, two cards each. Copy line for line; the faint numbers are your ruled lines.

WORD BANK · CONCEPT CARDS · scene 10 — C22

NUCLEAR THERMAL ROCKET = NTR · NTP

= a reactor heats hydrogen; a nozzle throws it

WHY fire caps exhaust at 4.5 km/s. A reactor needs no burning, so it can throw the lightest gas there is

\# ~2,700 K · 8.3 km/s · Isp ~850 s · 246,000 N (NERVA)

≠ RTG (electricity only) · nuclear pulse

→ $v_e \propto \sqrt{(T \div M)}$ · never flown

MOLECULAR MASS (M)

= how heavy one exhaust molecule is

WHY at the same heat, light molecules move faster, so light exhaust means high Isp

\# hydrogen 2 · steam 18 · exhaust speed ∝ √(T ÷ M)

≠ the propellant's total mass

→ why NTR doubles Isp while running cooler

WORD BANK · CONCEPT CARDS · scene 10 — C23

PAYLOAD

= the cargo that is the reason for the trip

WHY everything else (engines, tanks, propellant) exists only to move it; better Isp frees mass for it

\# Mars departure: 60% propellant at 450 s, 36% at 900 s

≠ the whole spacecraft

→ rocket equation · payload fraction

ENRICHMENT = LEU · HEU

= share of uranium that is splittable U-235

WHY more U-235 = smaller reactor, but bomb-grade material means heavy security and cost

\# natural 0.7% · LEU < 20% · 1960s engines 93%

≠ radioactivity level (different thing)

→ fission · reactor core · budget + schedule

C22: 11 + 10 lines of 22 · C23: 10 + 11 lines of 22

SEC 4 · LEC 4 · SAY IT OUT LOUD

Last page: prove it by talking

After the spread and the 3-line summary, with everything closed. Section 4 is complete after this, so the Section 4 talk is now in the app.

🗣 The friend's opening question

"A nuclear rocket? So it's basically a bomb going off behind the ship?"

✅ Three sentences you should be able to say

- No, it's a kettle with no fire. A reactor gets hot, you blow hydrogen through it, and the hot gas goes out an ordinary rocket nozzle.
- It gets about twice the mileage of the best chemical rocket, and the reason isn't that it is hotter. It is that the exhaust is plain hydrogen, the lightest gas, and light molecules fly faster.
- It worked on test stands in the 1960s but has never flown, because it is heavy, radioactive, expensive to test and politically hard.

❓ Follow-ups a friend might ask

- "Why would anyone want one?" (half the propellant for a Mars trip, or a faster trip and less radiation for the crew)
- "Could it get us to another star?" (no: tens of thousands of years even at its best)
- "Wasn't there one that did use bombs?" (Project Orion, 1958–1965, never built)
- "Is that what powers Voyager?" (no, that is an RTG: electricity only)
- "Which engine is best, then?" (none: chemical for push, electric for mileage, nuclear would sit in between)

📝 Your explanation (paste or dictate)

_ send it to me in chat; I'll put it here _

🎯 My read

solid on: _
fuzzy on: _
fix: _
→ question-bank cards: _

⚡ Section 4 done. Before Section 5: the Section 4 map page from memory, then one timed Section 4 talk from the app's Teach tab.